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Agentic AI

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Personal Habit Coach

Project Milestone II

System Overview

The Personal Habit Coach agent is a multi-agent AI system built with Streamlit and powered by the Groq API. The agents execute sequentially, relying on the previous agent's output to operate. There is an adjuster agent that operates outside of the sequential pipeline that revises the plan if needed. This agent is triggered by a separate user input. Habit data is stored across sessions with a local SQLite database. Users receive daily habit reminders through Gmail.

Agent Roles

Planner Agent

The planner agent is the starting point of the pipeline. It takes the user's desired goals and transforms them into a structured list of daily habits.

Tracker Agent

The tracker agent monitors the checked-off habits and calculates a completion percentage. It receives the full list of habits from the session state and live-checked habits in the checkboxes section.

Motivator Agent

The motivator agent generates a personalized encouraging message based on the user's completion percentage and self-reported feelings. This is the final step in the main pipeline. When this runs, the mood and percentage are saved to the SQLite database for history tracking.

Adjuster Agent

The adjuster agent is activated only when two conditions are met: the user selects "Struggling" as their mood and has granted the adjuster permission to modify their plan. If these conditions are met, it suggests a small change to make the habits more achievable.

Agent Communication

The agents communicate via the Streamlit session state, which serves as shared memory. The agents do not call each other; instead, they read the session state. The execution order is determined by the order in which the user clicks buttons.

Communication Flow:

Failure/Low-Quality Solution

1. Empty Input

If the user clicks Submit Goal without entering any text, a warning message pops up, and no API call is made. This prevents unnecessary API usage and guides the user.

2. API Rate Limits

If the Groq API encounters a rate limit, Streamlit will display an error message in the browser. Right now, the system uses the API's free tier, so usage is heavily restricted. For a class project, it is plenty, but on a production level, this would need to change.

3. Notification System

The notification system is a bit clunky. It requires the user's email address, the email app password, and a Task set up in Windows Task Scheduler. For now, the notification system works, but if this were a real product, this would need to be changed.

Agent System Architecture

